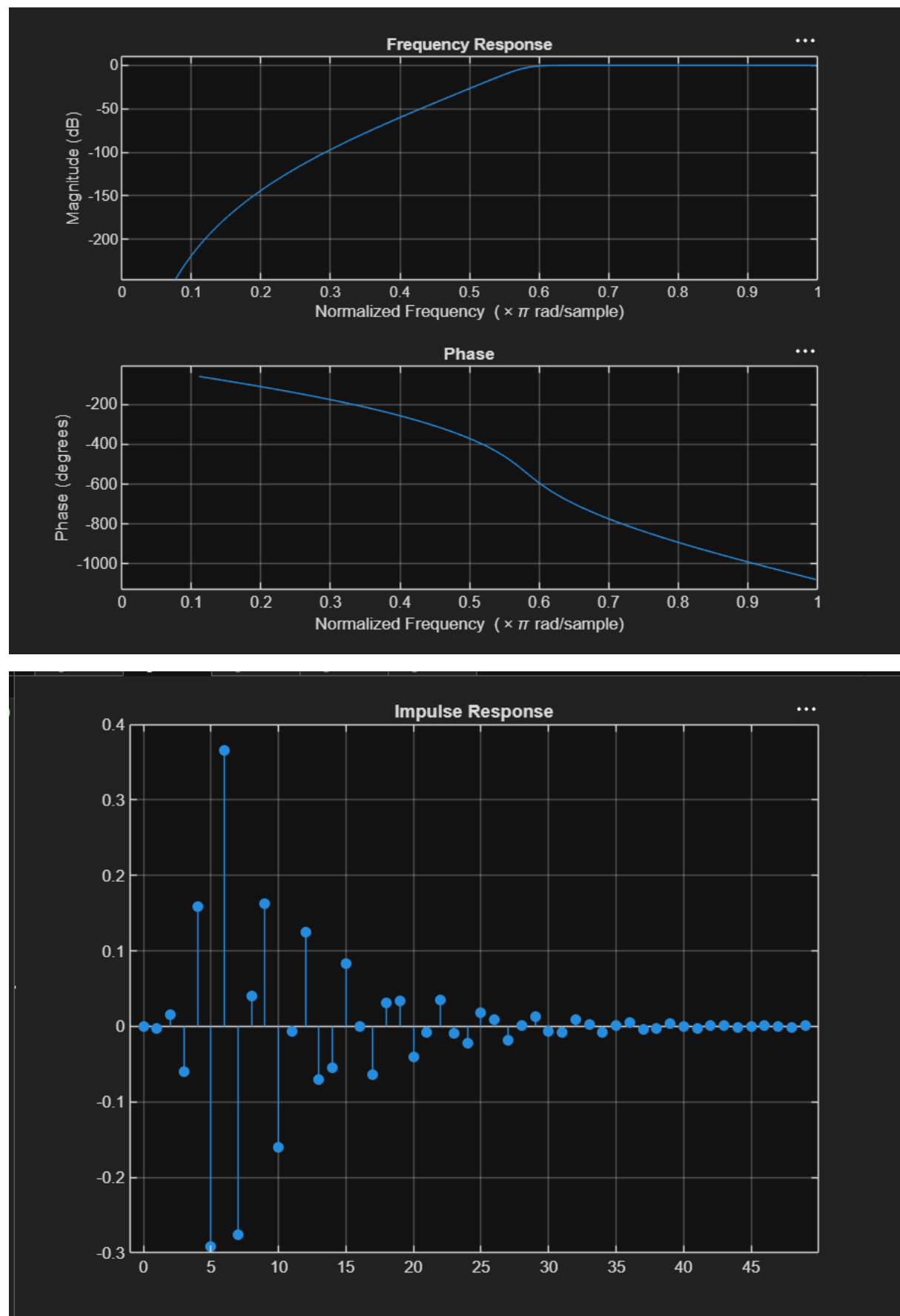
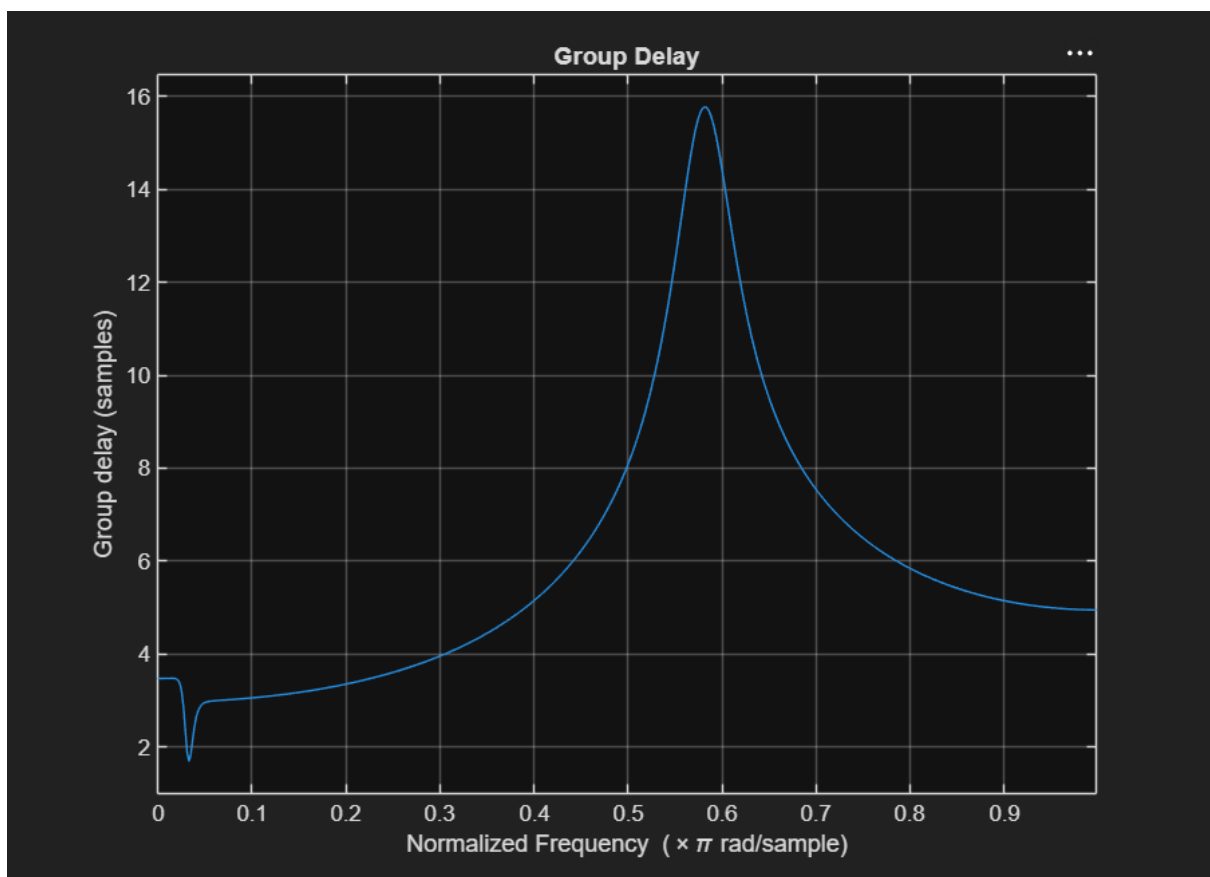
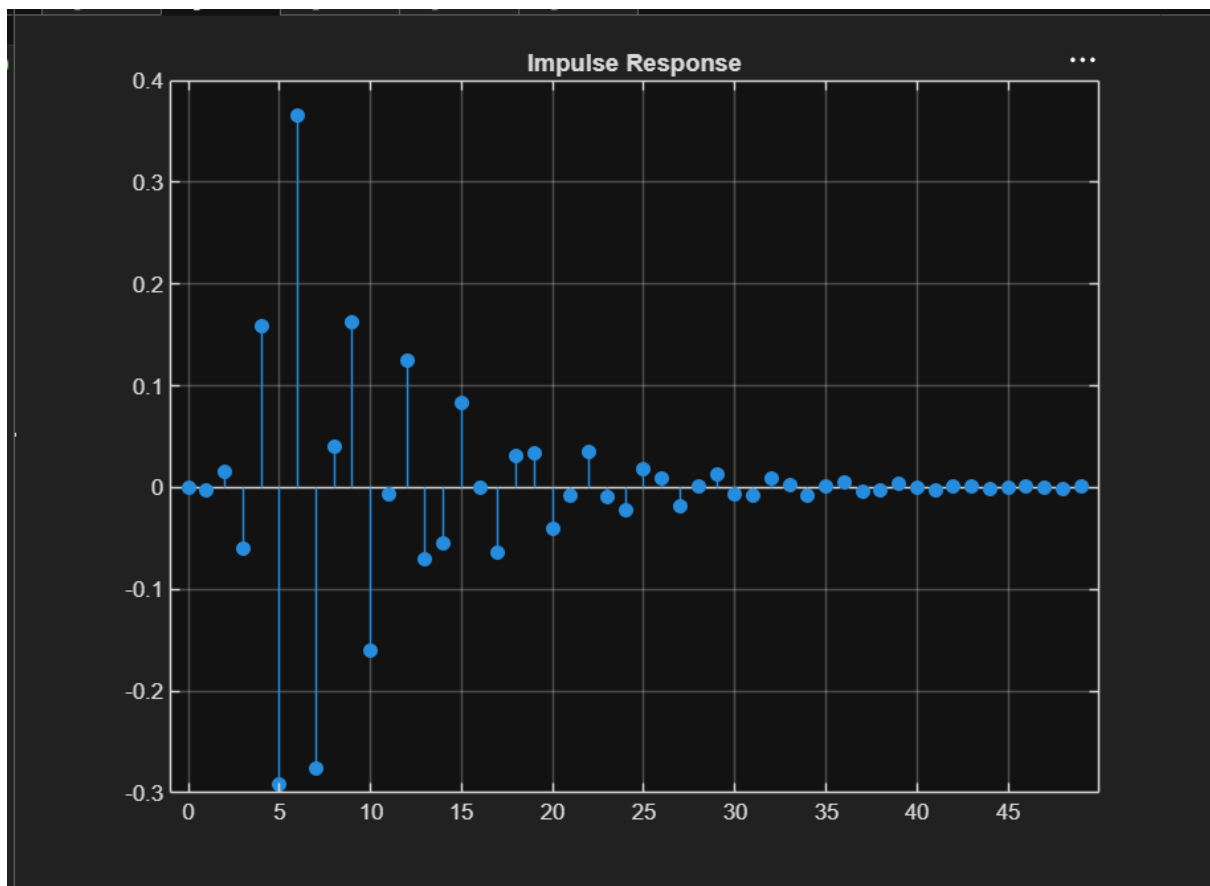
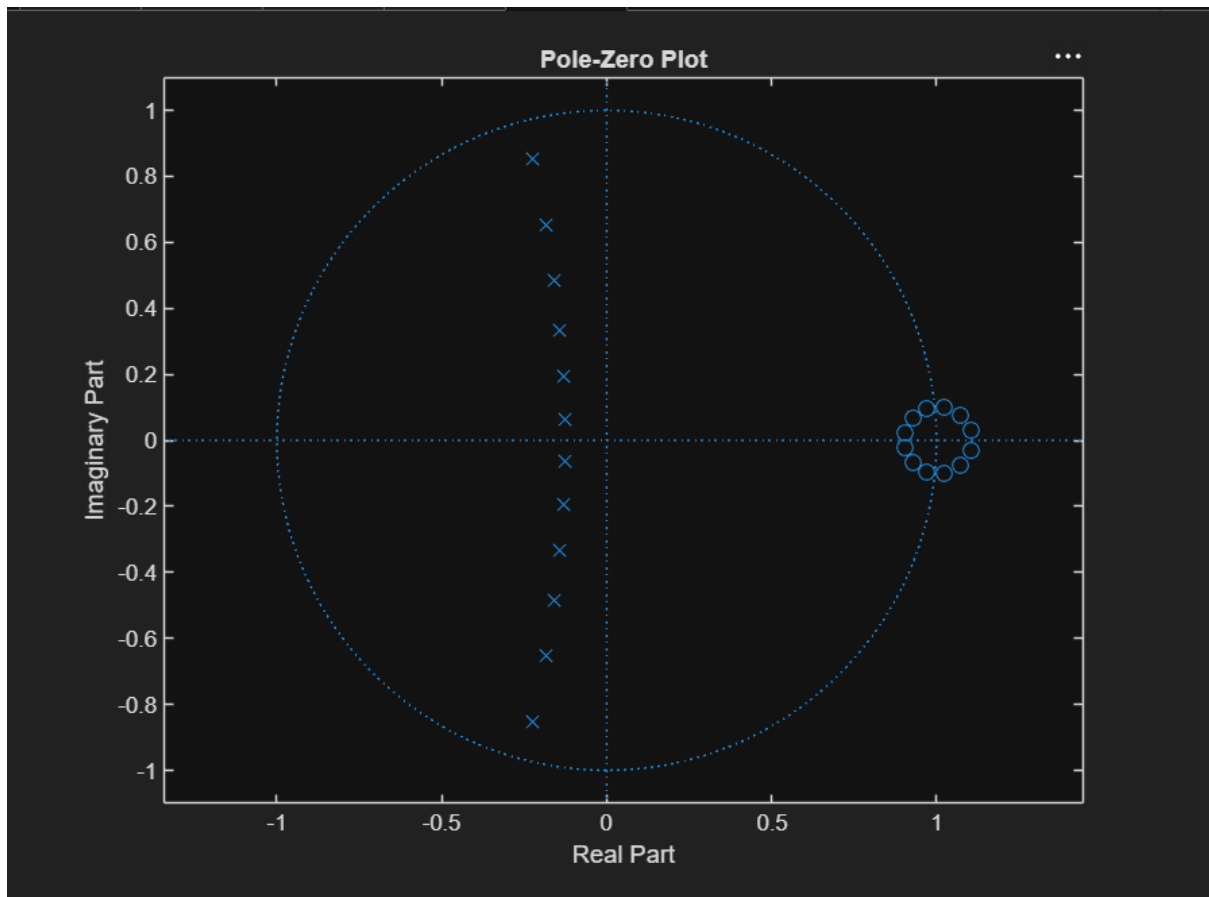


1.2 OUTPUT







PROGRAM

```
clc;
```

```
clear;
```

```
close all;
```

```
% EX NO: 1
```

```
% Butterworth IIR Filters
```

```
% DATE:
```

```
% 1.2 Design and Analysis of a Butterworth IIR High-Pass Filter Using MATLAB
```

```
% Specifications
```

```
Fs = 10000; % Sampling Frequency (Hz)
```

```
Fp = 3000; % Passband Frequency (Hz)
```

```
Fst = 2000; % Stopband Frequency (Hz)
```

```
Rp = 1; % Passband Ripple (dB)
```

```
Rs = 60;      % Stopband Attenuation (dB)
```

```
% Determine minimum filter order
```

```
[N, Wn] = buttord(Fp/(Fs/2), Fst/(Fs/2), Rp, Rs);
```

```
% Design Butterworth High-Pass Filter
```

```
[b, a] = butter(N, Wn, 'high');
```

```
% Frequency Response
```

```
figure;
```

```
freqz(b, a);
```

```
title('Frequency Response');
```

```
% Impulse Response
```

```
figure;
```

```
[h, n] = impz(b, a, 50);
```

```
stem(n, h, 'filled');
```

```
grid on;
```

```
title('Impulse Response');
```

```
% Step Response
```

```
figure;
```

```
stepz(b, a);
```

```
title('Step Response');
```

```
% Group Delay
```

```
figure;
```

```
grpdelay(b, a);
```

```
title('Group Delay');
```

```
% Pole-Zero Plot
```

```
figure;
```

```
zplane(b, a);  
title('Pole-Zero Plot');
```

```
% Display Results
```

```
fprintf('Minimum Filter Order (N) = %d\n', N);  
fprintf('Normalized Cutoff Frequency (Wn) = %.4f\n', Wn);  
fprintf('Cutoff Frequency = %.2f Hz\n', Wn*(Fs/2));
```